भारतीय प्रौद्योगिकी संस्थान कानपुर
Indian Institute of Technology Kanpur



AEROSPACE ENGINEERING

POST GRADUATE PROGRAMME

Website: https://www.iitk.ac.in/aero/

AEROSPACE ENGINEERING

Established in 1964, the Aerospace Engineering department of IIT Kanpur is a premier teaching and research centre today. The department specializes in the core domains of aerodynamics, flight mechanics, propulsion, structures and structural dynamics, and the multi-disciplinary areas such as computational mechanics, aero thermal sciences, and unmanned aerial systems. Over the years, the department has engaged with a large number of organizations including NAL, GTRE, HAL, DRDL, ADE, ARDE, ADA, ADRDE, ISRO, IAF, USAF, BARC, DMRL, DLJ, Boeing, Pratt and Whitney, Airbus and GE for scientific and engineering research. Among the world. class experimental and computational facilities, it includes the unique Flight Laboratory with three powered aircrafts and a moto glider along with a 880m long runway and DGCA-approved maintenance facilities. IIT Kanpur is proud to house the National Wind Tunnel Facility with a 3m x 2.25m test section which has supported leading national and international organizations for their testing needs.

The faculty expertise encompasses a wide range of domains and specializations which include but are not limited to computational and experimental subsonic, transonic, supersonic and hypersonic flows; design and optimization of the composite materials and structures for static, dynamic and impact loadings; multi scale material modelling; combustion-related computations and experiments; novel propulsion systems; space and satellite dynamics and control; guidance, navigation and control of aerospace vehicles; planetary mission design and systems engineering.

In addition to the long running premier MTech program in Aerospace Engineering, the department has launched a new MTech program in Unmanned Aerial Systems Engineering, supported by the Ministry of Electronics and Information Technology (MeitY), to give specialized training in Drone / Unmanned Aerial Systems. This is the first such program in India.



POST-GRADUATE PROGRAMMES OFFERED

M.Tech in Aerospace Engineering:

i. Aerodynamics
ii. Aero-Thermodynamics and Thermal Sciences
iii. Computational Mechanics
iv. Flight Mechanics
v. Propulsion
vi. Structures, Structural Dynamics and Aeroelasticity

MS(R) in Aerospace Engineering:

i. Aerodynamics
ii. Flight Mechanics
iii. Propulsion
iv. Structures, Structural Dynamics and Aeroelasticity

M.Tech in Unmanned Aerial Systems Engineering*:

i. Aeromechanics and Design
ii. Autonomy

Ph.D in Aerospace Engineering:

i. Aerodynamics
ii. Aero-Thermodynamics and Thermal Sciences
iii. Computational Mechanics
iv. Flight Mechanics
v. Propulsion
vi. Structures, Structural Dynamics and Aeroelasticity

** supported by Ministry of Electronics and Information Technology (MeitY)*



FACILITIES

- Aeromodelling Lab
- Flight Lab
- High Speed Aerodynamics Lab
- Low Speed Aerodynamics Lab
- National Wind Tunnel Facility
- Propulsion Lab
- Structures Lab



RESEARCH LABORATORIES

- Advanced Combustion and Acoustics Lab
- CFD Lab
- Combustion Lab
- Computational Propulsion Lab
- Fire Lab
- Flow Physics Computations and Analysis Lab
- Fluid Dynamics Lab
- Helicopter and VTOL Lab
- High Performance Computing Lab
- Hypersonics Experimental Aerodynamics Lab
- Jet-acoustics and Flow Control Lab
- Structural Analysis Lab
- Structures and Advanced Materials Characterization Lab
- System Dynamics and Control Lab
- Unsteady Aerodynamics Lab



FACULTY LIST

- **Abhishek, PhD (University of Maryland, College Park):** Rotorcraft aeromechanics, futuristic VTOL/STOL systems, aerial vehicle design, Unmanned Aerial Systems, Inverse flight dynamics and wind turbines.
- **A. Kushari, PhD (Georgia Tech):** Propulsion, Combustion, Liquid Atomization, Flow Control.
- **Ajay Vikram Singh, PhD (University of Maryland, College Park):** Combustion and Reacting Flows, Combustion Generated Functional Nanoparticles, Soot Formation and Oxidation, Fire Dynamics, Detonations and Explosions.
- **Alakesh C. Mandal, PhD (IISc Bangalore):** Experimental Aerodynamics, Flow instability and transition, Turbulent Shear flow.
- **Arnab Samanta, PhD (University of Illinois at Urbana-Champaign):** Fluid Mechanics, Aeroacoustics, Hydrodynamic Stability, Flow Control, Wave Motion.
- **Arun Kumar P, PhD (IIT Kanpur):** Subsonic and Supersonic jets, Flow control, Jet acoustics.
- **Ashish Tewari, PhD (University of Missouri at Rolla);** Flight Mechanics, Aeroservoelasticity, Space Dynamics and Control.
- **Ashoke De, PhD (Louisiana State University):** CFD, High Speed Flows, Flow-Acoustics Coupling, Fluid-Structure Interaction, Turbulence Modeling, Multiphase flows and Combustion, Energy Harvesting.
- **C. S. Upadhyay, PhD (Texas A&M University):** Adaptive and efficient FEM, continuum mechanics, continuum damage modeling of composites, phase-field modeling, topology optimization, super-wood and green composites, multi-functional structures.
- **D. Chaitanya Kumar Rao, PhD (IIT Kharagpur):** Evaporation and combustion of fuel droplets, Laser-induced evaporation and atomization of droplets, Pulse laser-induced atomization of droplets, Femtosecond laser-induced bubble dynamics.
- **D. P. Mishra, PhD (IISc Bangalore):** Combustion, CFD of Chemically Reacting Flows, Propulsion, Heat Transfer.
- **Debopam Das, PhD (IISc Bangalore):** Theoretical and Experimental Fluid Dynamics, Aeroacoustics, Instability & transition, Vortex Dynamics. Unsteady Aerodynamics, Bird's and Insect's Flight.
- **Dipak K. Giri, PhD (IIT Kharagpur):** Spacecraft Dynamics and Control, Space Explorations: On-orbit Servicing, Space Debris Removal, Orbital Infrastructure Design in LEO, MEO, GEO.
- **G. M. Kamath, PhD (University of Maryland, College Park):** Structural Health Monitoring, Composite Materials and Structures, Structural Dynamics, Condition Monitoring, Machine Learning, Aeroelasticity.
- **Kamal Poddar, PhD (UC San Diego):**Aerodynamics, Turbulence, Low and High Speed Flows.
- **Mohammed Ibrahim Sugarno, PhD (IISc, Bengaluru):** Experimental Hypersonic Aerothermodynamics, High Enthalpy Test Facilities, Shock Waves.
- **Navrose, PhD (IIT Kanpur):** Fluid Mechanics, Fluid-Structure Interaction, Optimization, Flow Stability and Control
- **Pradeep Moise, PhD (IISc, Bengaluru):**CFD, Vortex breakdown in swirling flows, Swirl in cardiac flows, tornadoes and wind turbines, Transonic buffet, Resolvent analysis.
- **Pritam Chakraborty, PhD (Ohio State University):**Mesoscale mechanics for plasticity, fatigue, creep and fracture; FEM; Multi-scale methods.
- **P. M. Mohite, PhD (IIT Kanpur):** Damage Mechanics of Laminated Composites, Composites, Finite Element Analysis, Inflatable and Deployable Space Structures, load bearing Radar Absorbing structures.

FACULTY LIST

- **Rachana Agrawal**, **PhD (Purdue University):** Space mission design; Planetary mission architectures; Astrodynamics; Space systems design and development; Systems engineering; Entry, descent, and landing (EDL); Small spacecraft technology.
- **Raghavendra P. Kukillaya**, **PhD (Princeton University):** Aircraft and Airship Flight Dynamics and Control, Systems Modelling, Simulation and Design, Optimal Control, Biomechanics.
- **Rajesh Kitey**, **PhD (Auburn University):** Fracture Mechanics, Composite Materials, Experimental Stress Analysis, Optical Metrology, Thin Films, Finite Element Method.
- **Rajesh Ranjan**, **PhD (JNCASR, Bengaluru):** CFD, Turbomachinery Flows, Applied Aerodynamics, Transitional and Turbulent Flows, Stability and Flow Control, HPC.
- **Rakesh Kumar**, **PhD (Penn State University):** Hypersonics, Rarefied Gas Dynamics, Microfluidics, Molecular Dynamics, Heat Transfer & Thermal Design.
- **Sanjay Kumar**, **PhD (Caltech):** Fluid Mechanics - bluff body wakes, shock-accelerated flows, shock waves, shock tubes.
- **Sanjay Mittal**, **PhD (University of Minnesota):** Unsteady Aerodynamics, CFD, FEM, Low speed Wind-Tunnel Testing, Bluff Body Flows, Shape Optimization, Sports Balls, Flows in air intakes.
- **Sathesh Mariappan**, **PhD (IIT Madras):** Combustion-driven oscillations, Machine learning methods to develop accurate low-order models, Optical flow diagnostics, and Acoustic measurements, Thermoacoustic machines, and Rotorcraft aerodynamics
- **Subrahmanyam Saderla**, **PhD (IIT Kanpur):** Online System Identification, UAV Design and Flight tests, Experimental Flight dynamics.
- **Tanmay Dutt Mathur**, **PhD (Penn State University):** Rotorcraft drive systems, Power gearbox for geared aero engines, wind turbines, and farm machinery, Design, testing, and reliability analysis of rotating machinery, Contact mechanics and tribology, Rotordynamics, Supercritical CO2 based turbomachinery, FEM, BEM, and multiscale methods for structural analysis
- **Tufan Kumar Guha**, **PhD (Florida State University):** Experimental Aerodynamics and Fluid Mechanics, Active and Passive Flow Control.
- **Vaibhav Arghode**, **PhD (University of Maryland, College Park):** Combustion, Heat Transfer, Fluid Mechanics, Experimental Methods, Computational Fluid Dynamics.



BROAD RESEARCH AREAS

- Acoustics & Aeroacoustics
- Aeroelasticity
- Aeromechanics
- Avionics
- CFD and High-performance Computing
- Composite Materials and Smart Structures
- Damage Modeling
- Design & Control
- Design and Dynamics of Autonomous Micro and Mini Air Vehicles
- Electric Propulsion
- Emissions
- Experimental Aerodynamics
- Experimental and Computational Combustion
- Flapping-Wing Flight Mechanics and Control
- Flight Testing
- Flow Control (High-speed & Low-speed)
- Flow Diagnostics
- Fluid-Structure Interactions
- Helicopter Theory (Dynamics & Aerodynamics)
- Hypersonic Aerodynamics
- Inflatable and Deployable Space Structures
- Instrumentation & Parameter Estimation
- Intake Aerodynamics
- Internal Flow Control (Active & Passive)
- Liquid Atomization and Spray Combustion
- Material Characterization
- Microfluidics
- Missile Guidance & Control
- Multiscale Modelling
- Planetary Exploration and Mission Engineering
- Plasticity, Fracture and Fatigue
- Space Dynamics
- Structural Design & Optimization
- Structural Health Monitoring
- Thrust Vectoring
- Transition and Turbulence
- Turbo Machinery
- Unmanned & Autonomous Air Vehicle
- Wind Energy and Design

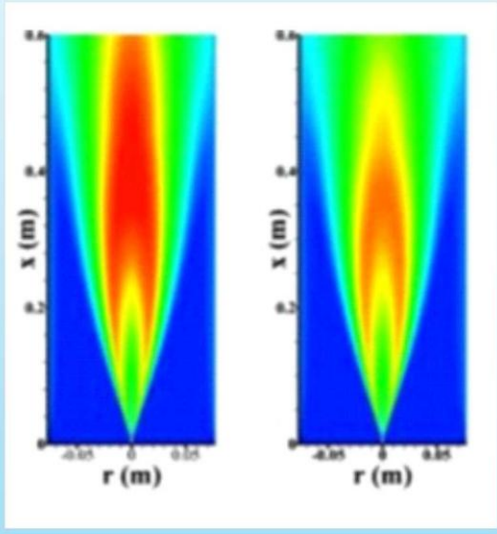
भारतीय प्रौद्योगिकी संस्थान कानपुर
Indian Institute of Technology Kanpur

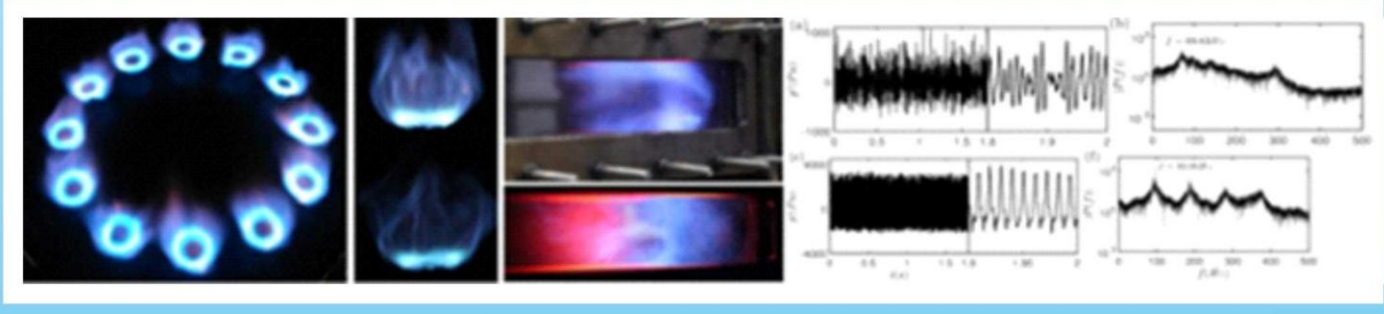
CONTACT

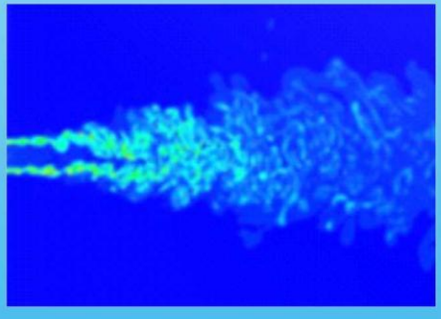
Prof. Arnab Samanta
Email: dpgc_ae@iitk.ac.in
Phone: 0512-259-2182

Mr. Vikas Kumar
Email: office_ae@iitk.ac.in
Phone: 0512-259-6704

Webpage: https://www.iitk.ac.in/aero/



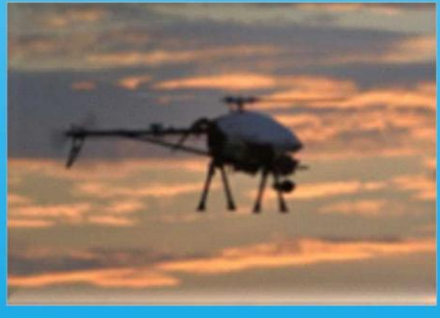
















भारतीय प्रौद्योगिकी संस्थान कानपुर
INDIAN INSTITUTE OF TECHNOLOGY KANPUR